UNIVERSITY OF ZAMBIA

SCHOOL OF HUMANITTIES AND SOCIAL SCIENCES
DEPARTMENT OF POPULATION STUDIES

DEM 1110: INTRODUCTION TO DEMOGRAPHY

APPLICATON OF SOCIAL AND ECONOMIC STATISTICS IN DEMOGRAPHIC RESEARCH

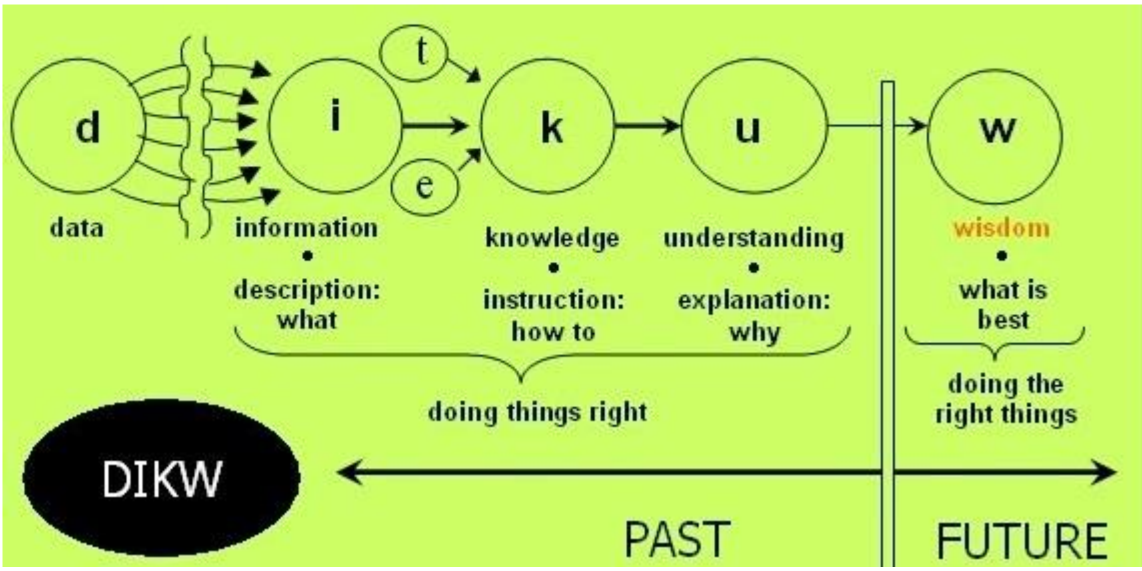
SCOPE OF SOCIAL AND ECONOMIC STATISTICS

Socioeconomic and demographic statistics are important for making comparison across countries on broad range of events/phenomena. Statistics are useful in assessing countries' economic performance, disease burden (incidence and prevalence, level of unemployment, crime etc. essentially we need statistics to measure, compare or count anything. In Demography data and information are the central focus. While these two concepts are used interchangeably there is a distinction. Data are the facts or details from which information is derived.

What Is Data?

- Data is a collection of numbers, symbols, sounds, actions, gestures pertaining to a phenomenon. Data may not tell a story until it is processed or analyzed.
- Data are a representation of reality that allows one to perform calculations meant to reveal information about the reality
- Data are the measures of properties or processes
- Data comes in different units of measurement

Data is essential in understanding how things happen, change or how the program is changing. Data are the basis from which information is generated, knowledge is acquired and wisdom is anchored. The Data, Information, Knowledge and Wisdom (DIKW) pyramid describes the data acquisition, its processing, retention and interpretations

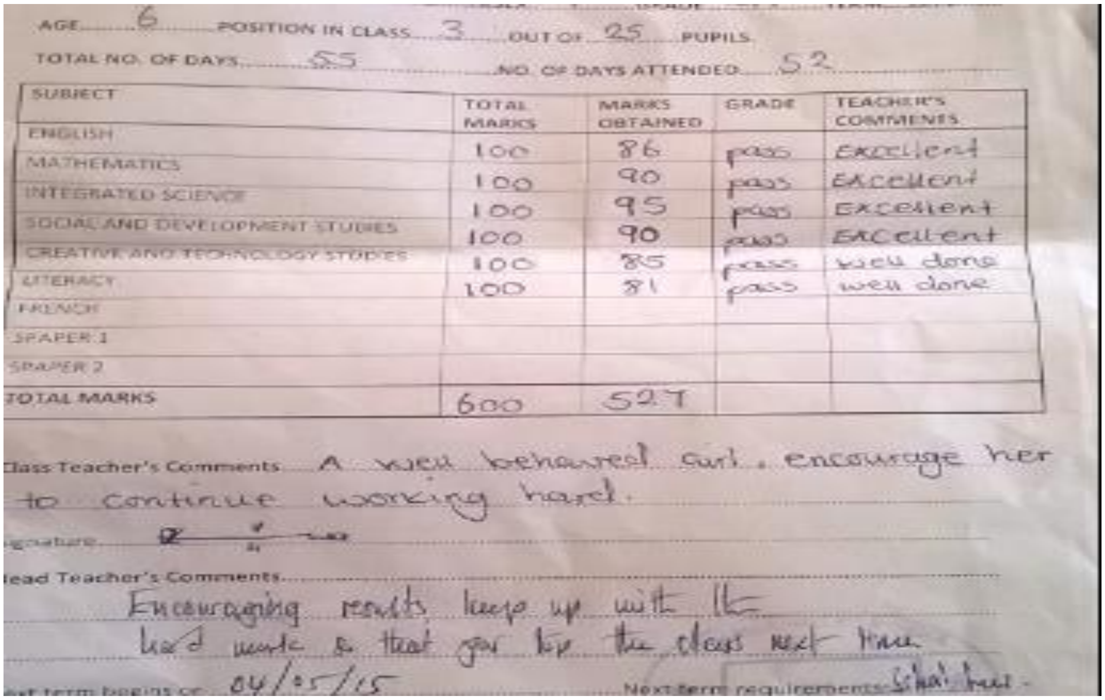


Therefore, Data are a mere collection of symbols or signs, details or facts about a phenomenon/ event. Information on the other hand are data endowed with context, meaning and purpose. Knowledge is a contextual information based on information from which patterns can be identified and be applied. Wisdom is the ability to increase effectiveness on the basis patterns identified. For example; think of the word "WifiPassword". The word alone is **data**. **Understanding** that it is a word is information. **Knowing** it is your wifi password is knowledge. And **using** it to access your wireless is wisdom. But everything starts with data.

What is Information?

This is processed data that provide meaning for observations made about a phenomenon of interest. These are data that are processed, organized, structured or presented in a specific context. A school report card is a basic example of information as it provides meaning about a pupil's performance in a class.

Report Card



AGE 6 POSITION IN CLASS 3 OUT OF 25 PUPILS

TOTAL NO. OF DAYS 55 NO. OF DAYS ATTENDED 52

SUBJECT	TOTAL MARKS	MARKS OBTAINED	GRADE	TEACHER'S COMMENTS
ENGLISH	100	86	pass	Excellent
MATHEMATICS	100	90	pass	Excellent
INTEGRATED SCIENCE	100	95	pass	Excellent
SOCIAL AND DEVELOPMENT STUDIES	100	90	pass	Excellent
CREATIVE AND TECHNOLOGY STUDIES	100	85	pass	well done
LITERACY	100	81	pass	well done
FRENCH				
SPAPER 1				
SPAPER 2				
TOTAL MARKS	600	527		

lass Teacher's Comments A well behaved girl. encourage her to continue working hard.

ignature

lead Teacher's Comments

Encouraging results keep up with the hard work & that you top the class next time

ext term begins on 04/05/15 Next term requirements School fees -

Types of Data

There are basically two types of data:

i. **Numeric**

These are data expressed as numbers/counts. Can be discrete or continuous

a. Discrete: these are counts that take whole numbers. E.g number of children, number of students in DEM 1110 class, population of Zambia, number of people not employed.

b. Continuous: there are data that are obtained by some form of measurement on an unbroken scale e.g age, height, weight, temperature, BMI

ii. **Categorical**

These are data that comes into distinctive group's e.g sex, marital status, HIV status, employment status.

Statistics: Statistics are a numerical value that describes some property of a phenomenon/event. These are numbers with context which may be social, economic or demographic.

Statistical methods are generally concerned with the study of how to **Collect, Organize, Analysis, Present and Interpret numerical information from data.** Statistical methods are required to ensure that data are interpreted correctly and that apparent relationships are meaningful. The processes of transforming data into information is called **Data Analysis.**

Statistics come in two forms:

(a) **Descriptive statistics –** Involves methods of organizing, and summarizing information from data

(b) **Inferential statistics** - Involves methods of using information from a sample to draw conclusion about the population.

NATURE OF SOCIO-ECONOMIC STATISTICS

Economic statistics: This may be defined as an historical record of economic activity which is capable of guiding the understanding of an economic system and at the same time capable of guiding the formulation of policy within the system. Quantitative information on manpower, production, distribution, transport, foreign trade, prices, employment, investments, national income and expenditures are examples of economic statistics.

Social statistics: This refers to data generated on the condition and quality of life of the people. Statistical information on household, education, health, public safety and population are examples of social statistics. Data are essential in making sound assessment of social and economic statistics.

COMMON CONCEPTS IN STATISTICS

Level: The amount of something. How much of a phenomenon of concern exists at a particular time. This can be social, economic or demographic e.g # of cholera cases, # of UNZA momas with Brazilian hair)

Pattern: This is the regular way in which an even happens or unfolds (what is the mortality rate).

Trend : This is the general direction in which an event or phenomenon is changing or manifests itself over a given period of time (how has been mortality rate).

USES OF ECONOMIC AND SOCIAL STATISTICS

1. Planning for national development
2. Construction of systems of national accounts
3. Construction of Economic Models
4. Policy formulation and decision making

PROBLEM OF COLLECTING ECONOMIC AND SOCIAL STATISTICS IN ZAMBIA

1. Conceptual problem
2. Problem in the statistical system

- ✓ Shortage of well-qualified statistical manpower
- ✓ Inadequate coordination, cooperation and collaboration among major producers of statistics in the country
- ✓ Inadequate funding of the statistical agencies
- ✓ Administrative bureaucracy and red tapism

3. Problem in Society

- ✓ Lack of statistical awareness
- ✓ Illiteracy/ innumeracy
- ✓ Cultural /religious problems
- ✓ Language problem
- ✓ Poor social facilities

MAJOR PRODUCERS OF SOCIO- ECONOMIC STATISTICS IN ZAMBIA

1. CENTRAL STATISTICAL OFFICE

The CSO is the main national agency responsible for the development and management of official statistics in the country. It is the authoritative source and custodian of official statistics in the country. CSO is a department under the Ministry of Finance and National Planning established by an act of parliament, Cap 127 of the laws of Zambia. Central statistical Office has several subject matter divisions such as;

- ✓ Economic Statistics
- ✓ Social statistics
- ✓ Agriculture statistics
- ✓ Labor statistics
- ✓ Information , Research and Dissemination

Functions of the Central statistical Office

- ✓ To coordinate the national statistical system
- ✓ To advise government, ministries and other agencies on all matters related to statistical development
- ✓ To develop and promote use of statistical standards and appropriate methodologies in the system
- ✓ To collect, compile, analyze, interpret and disseminate information alone or in collaboration with other agencies, both government and non-governmental agencies
- ✓ To develop and maintain a comprehensive national data bank by encouraging unit of line ministries and agencies develop their sectoral data bank and forward to CSO
- ✓ To provide a focal point of contact with international agencies on statistical matters
- ✓ To carry out all other functions relating to statistics as might be assigned by government

Statistical Activities of CSO

- ✓ Conduct the National Census
- ✓ Conduct the Demographic and Health surveys (ZDHS)
- ✓ Living Conditions and Monitoring survey (LCMS)
- ✓ Economic Census (EC)
- ✓ Agriculture Survey
- ✓ Labor force survey
- ✓ Zambia HIV/AIDS Impact Assessment (ZAMPHIA)
- ✓ Monthly Food Baskets

2. CENTRAL BANK OF ZAMBIA (BANK OF ZAMBIA)

The Bank of Zambia publishes banking, financial, foreign trade statistics etc through its;

- ✓ Annual Report and Statement of Account
- ✓ Economic and Financial Review
- ✓ Monthly Report
- ✓ Principal economic and financial indicators
- ✓ Statistical Bulletins, e.t.c

3. Other Sources

- ✓ Line Ministries and government agencies
- ✓ Bilateral organizations (World Bank, UN, WHO etc)
- ✓ UN Department of Economic and Social Affairs, Population Division.

  http://www.un.org/esa/population/unpop.htm

- ✓ UN Demographic year book
- ✓ UNDATA
- ✓ Web bases sources (US. Census Bureau, UN stats, etc)

- World Health Organization (WHO).
  http://www.who.int/en.
- Human Mortality Database.
  http://www.mortality.org.
- United Nations Population Fund
  http://www.unfpa.org/public/
- OECD
  The Organisation for Economic Co-operation and Development
  http://www.oecd.org/home/0,2987,en_2649_201185_1_1_1_1_1,00.html\

DATA COLLECTION TYPES

- Data collection is a term used to describe a process of preparing and collecting data
- It is a systematic gathering of data for a particular purpose from various sources.
- Data are the basic inputs to any decision making.
- The raw material for socio economic investigation and analysis is data.
- The type of data collected for a given purpose may be **Primary** or **Secondary**

Purpose of Collecting Data

- Obtain information
- Keep record
- Make decision about important issues
- To pass information to others

Primary Data

This is data collected at first hand for a specific purpose. The basic sources of primary data are Interviews, observation and administrative processes such as medical examinations, voter registration, national registration cards other sources include Census, surveys, vital registrations etc. Primary data are collected where

a. The needed information does not exist elsewhere

b. The needed information exist but is not reliable

c. Collecting the information at first hand is only way such information can be obtained

Advantages of Primary Data

- Targeted issues are addressed
- Data interpretation is better
- Proprietary issues
- Address specific issues
- Greater control

Disadvantages

- Cost
- Time consuming

- Inaccurate feed-back
- More resources are required
- Differences in conceptual definitions

Secondary Data

These are data which already exist and may be adapted for use in the current survey. Such data are collected originally for another purpose. Secondary data can be sourced from publications and records of governments and non-government organizations, journals of universities and research institutes, media, organization and administrative records.

For secondary data to be used with reasonable degree of confidence, the validity of such data must be assessed. This involves checking for the following;

- ✓ The source of the data
- ✓ The purpose of which it was collected
- ✓ The method of data collection used
- ✓ Definition of terms used
- ✓ Coverage and changes overtime, if any
- ✓ Method of analysis

Advantages of Secondary Data

- Faster
- Less Expensive
- Less activities (Field trip, Survey etc.)

Disadvantages

- Not easily available
- Not adequate
- May not meet the needs of researcher
- Outdated information
- Variation in definition
- Inaccurate or bias

Internal and External Secondary Data

a. Internal Data

This is data collected by one’s own organization. Such data may be extracted from records such as the hospital register books, police records on crime, company’s ledger books, personnel records, sales records, stock records e.t.c.

b. External Data

These are data collected by one organization and being used by another organization. Libraries, books, magazines, newspapers, journal articles, internet are examples of external sources of secondary data

SOURCES OF SOCIAL AND ECONOMIC STATISTICS

1. CENSUS

This is the total process of collecting, compiling and publishing demographic, economic and social data pertaining at a specified time to all persons in a given country or delimited territory.

Characteristics of a census

1. Universality- every unit or individual must be counted
2. International comparability- Should be conducted using the standards and principles of all census worldwide
3. Sponsorship- Because it is an expensive venture
4. Must focus on an individual (information collected pertains to each and every individual)
5. Periodicity- it must have a specific period of time e.g 5 or 10 years
6. Simultaneity- it should be taking place at the same time with other countries for the purpose of comparisons.
7. Legality – it must have a legal basis. E.g the CSO is mandated by an act of parliament to conduct a census.

There are two methods of counting in a census:

De-facto and De-jure

De-facto

This counts people from where they are found (People who spent a night at a household) regardless of whether they stay there or not. This method excludes;

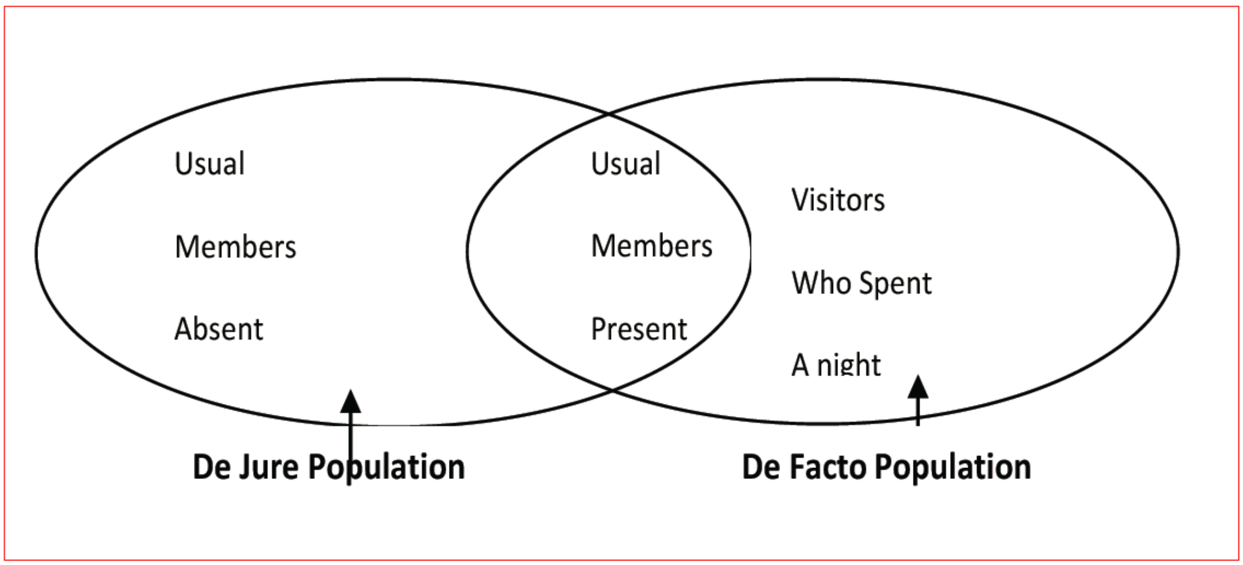
- ✓ Foreign diplomatic personal accredited to Zambia
- ✓ Zambian nationals accredited to foreign embassies and their family members who live with them abroad

- ✓ Zambian migrant workers and students in foreign countries who might not be in the country at the time of the census

De-jure

This counts the usual household members present and usual household members temporarily absent at the time of the census. In de-jure methods, populations in institutions such as hospitals, universities/colleges, prisons, schools are counted as members of their usual household

Figure 2.1: Diagrammatic Presentation of the De Facto and the De Jure Populations



Advantages and disadvantages of De-facto and De-jure Method of Counting

The De-facto count tends to overestimate (inflate) the actual size of the population whereas the De-jure tends to underestimate the population size.

Methods of Enumeration

There are two methods of getting information from a respondent

a. Preparing a questionnaire: The enumerator trained by a statistical organization (CSO) gives a questionnaire to the oldest member of the household who complete the questionnaire (answer the questions in the questionnaire)

b. Recruiting a number of people (enumerators) who go round household by household collecting data

Disadvantages of preparing a questionnaire

- ✓ Some respondents are illiterate
- ✓ Some questions might need clarification, thus may be errors in the answering some questions
- ✓ High non-response errors

Advantages of preparing a questionnaire

- ✓ The respondents has enough time to respond and check or verify any information (checking family archives)
- ✓ Guarantees a lot of confidentiality as some people may be shy to respond to question face to face with an enumerator
- ✓ The method is less costly

Advantages of Census

- ✓ It is a good source of information as it covers everybody and doesn't suffer from sampling errors
- ✓ The time between the censuses is adequate as people can prepare for the census adequately
- ✓ Provides information on cross-cutting issues such Housing, population, agriculture, economic performance etc

Disadvantages of Census

- ✓ It is a very costly undertaking as it covers everybody
- ✓ It has inherent problems of coverage errors as some areas cannot be accessed
- ✓ The census lacks detail on the indicators it covers (descriptive in nature)
- ✓ Very costly

2. SURVEYS

Sample surveys are among the major sources of socio-economic and demographic data in Zambia. Sample surveys acknowledge the importance of censuses. However, censuses are conducted at large intervals e,g ten years and they tend to collect information with less detail or depth.

A sample survey is a scientific data collection process where only a proportion of the population is involved.

Approaches of Capturing Information from Respondents

There are two methods

a. **Retrospective:** Focuses on events that have already occurred/happened
b. **Prospective:** Focuses on events as they are happening. E,g participating in class, the spread of ebola virus etc

Advantages of a Survey

- ✓ The sample surveys compared to the census provide a cheaper alternative for timely data and more relevant and convenient social, economic and demographic data pertaining to conditions under which people live, their wellbeing, activities in which they engage in such as demographic characteristics, cultural factors which influence behavior as well as socio-economic changes.
- ✓ Surveys have a more flexible method of data collection, making it an excellent choice for meeting data users' needs.
- ✓ It collects data at a greater depth/detail due to larger time given for interviews and smaller interview workloads (detailed and specific subject matter).
- ✓ In reality not all the data needs of a country can be met through census undertaking. Therefore sample surveys provide a mechanism for meeting the additional and emerging needs on a continuous basis.
- ✓ Social, economic and demographic characteristics that change frequently in some instances or areas such as fertility, motility and migration, employment can be captured
- ✓ Easy to observe events over time in a survey than a census

Disadvantages

- ✓ Prone to sampling errors
- ✓ Prone to abuse
- ✓ Poor response rate because they are not legally bidding as the census
- ✓ The size of the sample only ensures that reliable statistics can be shown to a very limited geographic detail
- ✓ Coverage is only extended to a limited geographical area

Link between Census and Surveys

- ✓ Census data/statistics serve as benchmark for analyzing and evaluating survey data
- ✓ Census is often used as a sampling frame for selecting the population to be surveyed in a census (means of selecting a specific survey population group)
- ✓ Surveys have been used to determine the amount of error in a census (Post enumeration survey)

SURVEY PLANNING

A. Stages of Planning

1. Survey Planning
    a. General Planning of the survey program
    b. Selecting and specification of the subject matter and preparation of the tabulation plans
    c. Developing of the survey design
        i. Decision on type survey framework
        ii. Scheduling and timing consideration
        iii. Decision on population coverage
    d. Developing of budget estimates and a calendar of operations
2. Survey Preparation
    a. Design of survey samples
    b. Developing of cartographic materials
    c. Developing of survey procedures
        i. Determination of data collection procedures
        ii. Design of a survey questionnaire
        iii. Pre-testing of questionnaire and sample procedures
        iv. Preparation and training materials for field work
        v. Decision on selecting of survey respondents
        vi. Decision on respondents compensation
        vii. Development pf a quality control system for the survey operation
        viii. Decision on data processing, methodology such as the appropriate mixture of clerical and computer operation and the extent of manual coding and editing
        ix. Institution of an appropriate publishing and public relations

B. General Planning

Planning is usually carried out by a relatively small group of senior staff representing subject matter, technical and administrative backgrounds. At this stage several factors have to be taken into consideration.

i. Selection of the subject matter and other priority areas
ii. Establish a general time table which is logically coordinated
iii. Survey design, this is highly technical and requires detailed study population. The group should sample most appropriate design taking into account
    a. Country's physical features
    b. Population distribution
    c. Availability of transport
    d. The subject matter should be investigable
iv. Staffing for various aspects of the program , such as
    a. Recruitment
    b. Training
    c. Supervision
v. Other resource requirements
    a. Space
    b. Equipment
    c. Basic facilities
    d. Budgetary resources

C. Subject Matter

1. Selection and specification of data requirements
    i. This depends on the needs such as government statistical needs
    ii. Subject matter specialists must translate the needs of users into detailed specification of survey use
    iii. The subject specialists must determine whether the proposed data can be obtained reasonably from households survey respondents
2. Organization of Subjects
    i. Check which questions, topics can suitably be combined and which should be taken separately
    ii. Important approach is to identify "Core Item" e.g demographic characteristics- age, sex, nationality, marital status
    iii. Generally consists of standardized groups on particular topics which can rotate from survey to survey. E.g in ZDHS there are sections such as Child Health, Contraception/ Family Planning, nutrition. The core provides continuously updated body of basic statistics. It also permits the use of a standard set of questions across the various subject fields
    iv. The combination and inclusion of subjects into a survey is subject to;
        a. Budget

b. Administration
c. Needs

v. When combining check and make consideration as to which ones archives or produces required dividends
vi. How logical are they going to be to respondents (you cannot combine fertility and crime or irrigation in one subject.

Caution to Subject Matter Combination

i. Over loading the operation (both the interviewer and respondents)
ii. Danger of over-burdening the data processing and analysis phase

Comparability of data with other sources

Important for subject matter staff;

i. Compare with census
ii. Compare with previous surveys
iii. International recommendations or as used in other countries
iv. Questionnaire designs is also important

3. VITAL STATISTICS REGISTRATION SYSTEM (VRS)

The vital registration system involves the continuous and permanent recording of vital events pertaining to births, deaths, marriages, divorces, legal separations, annulments etc. in developing countries the system is still under problems. The information is collected by the civic authorities such as the city, district, and municipal councils.

The vital registration system gives certificates to the occupancy of events such as Birth, death and marriage certificates.

USES OF CERTIFICATES

a. **Birth Certificates (issued by the local authority upon child's birth)**
    - ✓ Prove of identify
    - ✓ Claiming benefits (Inheritance) (in case of parents dying)
    - ✓ Verifying Childs age
    - ✓ Verifying parentage and relationship of the child

b. **Death Certificates (issued by the attending doctor and the burial permit is issued by the civic authority)**
    - ✓ Used in pension schemes to prove an individual's death so that the surviving relatives can claim benefits
    - ✓ Used to transport bodies from one place to another for burial
    - ✓ Used by social welfare authorities
    - ✓ Used to estimates birth rates

c. **Marriage Certificates (issued by the local authorities and other institutions)**
    - ✓ Proof of marriage
    - ✓ Claiming of benefits when spouse dies, divorces
    - ✓ Used to observe marriage and fertility patterns

Vital statistics are used to estimate Crude Birth Rate (CBR), crude Death Rate (CDR) when the system is well established.

Types of Registration

Broadly there are two types of registration systems of vital events

i. **Passive System:**

    This approach relies on the good will and cooperation of the public (individuals) to report events as they occur. This implies that registration personnel wait until informants (public/individuals) report the events)

ii. Active System

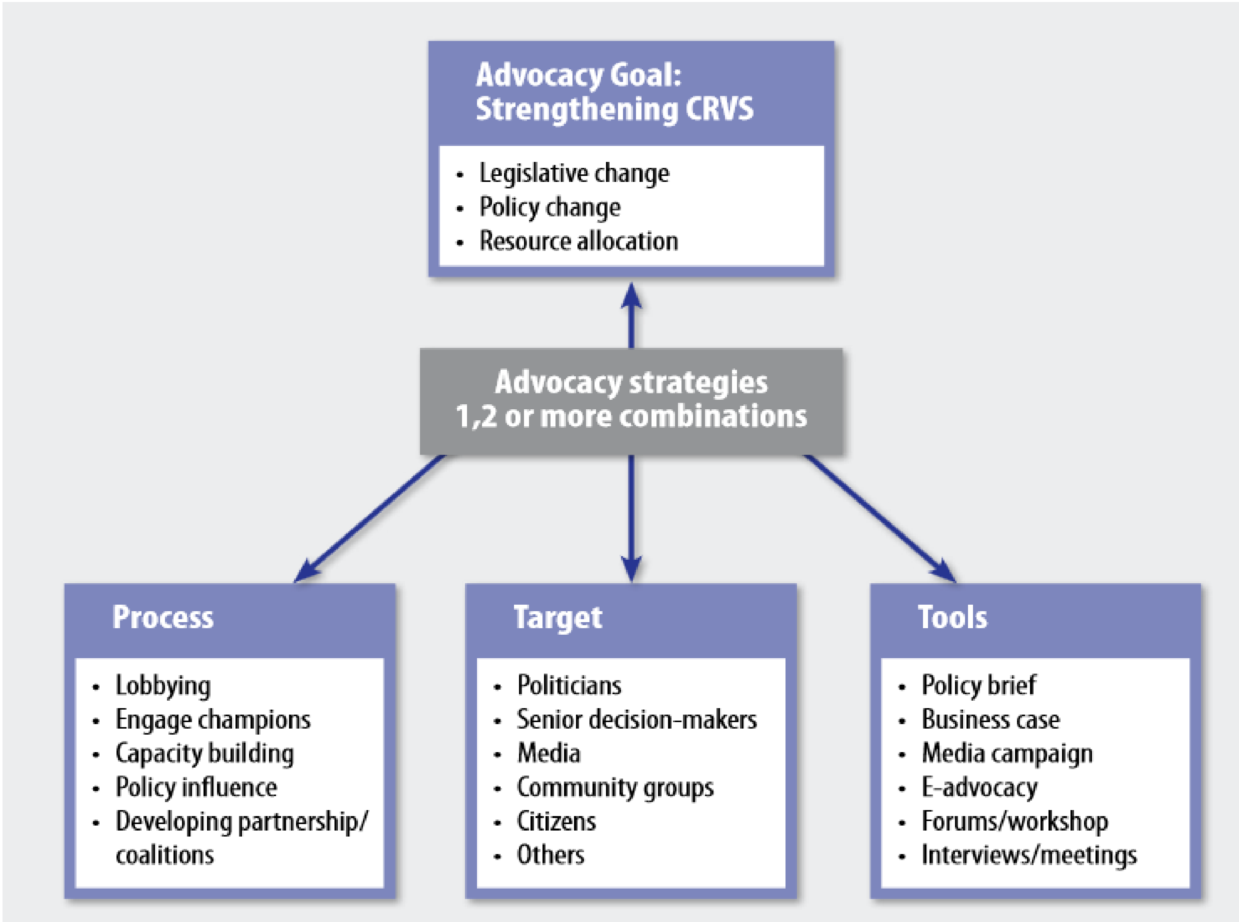
Under this approach, the registration institution employs registration personnel who go round to canvass the vital events that occurred rather than wait for informants to do so. As part of the system, chiefs, teachers, health workers, midwives, religious leaders and other local government functionalities could be used as rotifers

Problems of VRS

i. Underfinance from the part of government (lack of funds) to support the recording of vital events. Reliance has been placed on individuals to report and have vital events recorded
ii. It's not mandatory as such people do not see the utility of registering vital events such as births, deaths etc making the whole registration of vital events fragile
iii. Due to high Illiteracy levels, people do not see the need of registering (especially in rural areas
iv. Lack of infrastructure (both systems and physical infrastructure) to support the registration of vital events
v. Lack of administrative structures at lower levels (ward, constituency)
vi. Low status and lack of expertise
vii. The time lapse between the occurrence of a vital event and registration make most vital statistics unreliable and fraud
viii. Problems of rural areas

IMPROVING VITAL REGISTRATION SYSTEM FOR NATIONAL ECONOMIC PLANNING AND SOCIAL DEVELOPMENT

i. Strengthening the legislation system governing the registration of vital events
ii. Making the registration of vital events compulsory and continuous
iii. Strengthening of infrastructure through local government structures to the lowest organ of administration
iv. Strengthen advocacy on registration of vital events



v. Build and strengthen working partnerships with local and international organisations (churches, chiefs, headmens etc)

4. POPULATION REGISTER

The Population Register is a comprehensive identity database that is maintained on a continuous basis and contains all people in a country and their characteristics.
Such as;

i. Date of birth
ii. occurrence of any event such marriage, divorce, place of occurrence, characteristics of siblings or children and that of the deceased
iii. Characteristics of the parents of the new born
iv. Cause of death in-case of the deceased
v. Migrations (in and out)

If a population register is well maintained, it gives all the details and they might not be need for a census. Some countries such as Austria, Denmark, Finland, Israel, Japan, The Netherlands, Norway and Sweden have removed the need for censuses by adopting a system whereby population statistics are continuously updated based on information recorded in administrative registers.

Population registers are designed in such a way that in each locality e.g the council, city or town there will be a listing process (register) where all the characteristic of people are entered. Each person is assigned a unique identifying number which enables information across different registers (i.e. birth, death, marriage etc.) to be linked and for central records to be updated. The linking of records means that as well as providing population statistics, the registers can also be used for longitudinal data which follows an individual across the life course.

The procedure is very simple but sitting on a very complex system;
When someone is born a card is opened and the information is updated as the person progresses in life. When one moves, he/she is registered as a migrant and the both the place of departure and arrival are notified. When one dies, the card is closed to indicate the terminal end of the person life.
In Zambia, they were what were called Village Registers which had all the characteristics of the people in the village. The Village Registers where deposited at the Chief's palace. When someone is born, information was recorded; when one moves he/she was given a letter addressed to the place of destination.

Advantages

i. The continuous nature of data collection (or supply) for population registers means that the information available is current, whereas information from decennial censuses may be many years out of date.

ii. Population registers can be updated almost instantaneously, meaning that there is no long waiting period for the data to be collated and analyzed as there is with decennial censuses.

iii. There is a potential for linked records, meaning that (anonymised) individuals can be followed through the life course for research purposes

Limitations of Population Registers

i. Very expensive to maintain as it requires setting up of structures (both systems and physical almost everywhere)
ii. Its passive in nature
iii. Does not work effectively in countries with high illiterate population
iv. The type of data available is determined by the nature of the administrative departments collecting it. On a census, additional questions can be easily added to give data on a specific area of interest.

v. When first established, the quality across different departments or regions may be of uneven quality.

vi. Events occurring before the start of registration, or occurring while the individual was abroad, may be excluded.

vii. The fact that the record is permanent, follows you through life, and is linked between departments, may be unacceptable to some with concerns of a "Big Brother" style society. There are also concerns regarding the potential for data leaks.

EXAMPLES OF SOCIAL ECONOMIC STATISTICS

1. HEALTH STATISTICS

International demands and the need by national governments to provide or improve health conditions has led to search for better data on all aspects of health. Hence, the need to keep health statistics for Zambia, the Ministry of health keeps and updates what is called Health Information Management system (HMIS) based on data generated at the facility level. Other statistical health information systems include the National AIDS Council e-mapping, SmartCare.

The health conditions in a country are reflected by;

a. Morbidity and mortality in a population
b. Fecundity and fertility

Morbidity and mortality are affected by the social, economic, political conditions as well as modernization such as medical technology, access to health services

Health statistics are mainly recorded in terms of mortality, morbidity and fertility.

SOURCES OF HEALTH STATISTICS

- ✓ Hospital records: these are mainly hospital records or patient registers such HTC register, maternity register etc.
- ✓ Surveys (ZDHS), Sexual behavior survey (SBS)
- ✓ WHO records
- ✓ NGO

USES OF MORBIDITY STATISTICS

- ✓ Control epidemics; administrative action for identification and isolation of infectious diseases
- ✓ Medical intelligent systems
- ✓ Research
- ✓ Program development and implementation

TYPES OF HEALTH DATA

Population data:

- ✓ Population distribution (rural urban)
- ✓ Age-sex to measure distribution of vital statistics i.e mortality, fertility
- ✓ Literacy (education plays an important role on health)
- ✓ Economically active population (studies in occupation health hazards)

HEALTH PROBLEMS DATA

- ✓ The reporting of such data is very slow because sickness is not necessary a vital event and there is no statutory obligation on reporting.
- ✓ Much of the health statistics document diseases, incidence and prevalence of illness
- ✓ Data on the utilization of health services and medical care facilities
- ✓ Immunization status of the population
- ✓ Information on related social economic status

USES OF HEALTH STATISTICS

- ✓ To evaluate the health status of the population
- ✓ To investigate factors influencing health
- ✓ To evaluate the effectiveness of health measures and assessment of future needs
- ✓ To study the utilization of health services

HEALTH SERVICES RESOURCES DATA (INPUT)

This deals primarily with the resources available or health services and the extent to which they cover the population i.e by way of access.

The statistics required cover:

- ✓ Number of health centers, hospitals, clinics and their location

- ✓ Number of hospital beds
- ✓ Health manpower (doctors, nurses, midwifes, clinical officers)
- ✓ Training rates of health manpower
- ✓ Total employment in the health sector

USES OF HEALTH RESOURCES DATA

- ✓ Assess present/future health requirement in terms of services and facilities
- ✓ Assess the needs of various social/economic groups
- ✓ Evaluate the effectiveness of health services
- ✓ Evaluate the effects of internal ,migration in health services

HEALTH SERVICES (OUTPUT) ACHIEVEMENTS

These include;

- ✓ Hospital attendance, outpatients' attendances, deliveries attended by qualified personnel, immunizations, inpatient disease patterns. This data is collected by means of cards designed by health departments

HEALTH SERVICES COSTS

This data is all involving as it involves

- ✓ Total expenditure on health by health agencies
- ✓ Revenue collected
- ✓ Costs in and out of patients
- ✓ Costs of health training
- ✓ Pharmaceuticals
- ✓ Etc

SIMPLE METHODS OF ANALYZING HEALTH DATA

Average length of Stay=Patient- days/Admissions

Occupancy Rate= (Patient-Days/ bed-Days)*100

2. LABOR STATISTICS

SCOPE

Generally they are part of the important national statistical program and closely interrelated with statistics in other fields such as education, health and production.

Include information on the size, structure of the labor force, numbers of people who work (production, service delivery, characteristics or workers, their jobs, their employer, income, type of payment received (regular or part-time, hours worked per day, occupational exposures, and related injuries, type of contracts, unionized or unionized workers)

Employment/ labor status is a highly contentious issue of our time, a country where only about 750,000 people out of the about 13 million are in formal employment.

Generally, labor statistics provide a basis for assessing the size of the labor market, the level of unemployment, how serious the problem of unemployment can possibly be as well as the utilization of the labor force thus identifying the labor force gaps, deficits and surpluses

DEFINITIONS AND MEASUREMENTS

a. **Economically Active:** This consist of a population which is either working or looking for work between the age of 15-65

b. **Economically Inactive:** These include those who are too young (below 14), too old to work, physically challenged as well as students

c. **Employer:** Operate their own businesses and engage others

   Characteristics for Employer:

   i. Production or service Delivery
   ii. Institutional sector (private or public)
   iii. Demand for labor
   iv. Cost of employing labor

d. **Unpaid Worker:** These are a group of people who work but are not paid/salaried workers, often family members e.g, house wife, children or generally family members

e. **Employed:** Wage earner regularly working at least half of the time

f. **Underemployed:** irregularly working, daily workers with very low number of hours, or low productivity job compared to the profession/different from the profession

g. **Quasi-Employment:** Own account workers in non-productive engagement e.g shoe cleaners, street vendors, car watchers

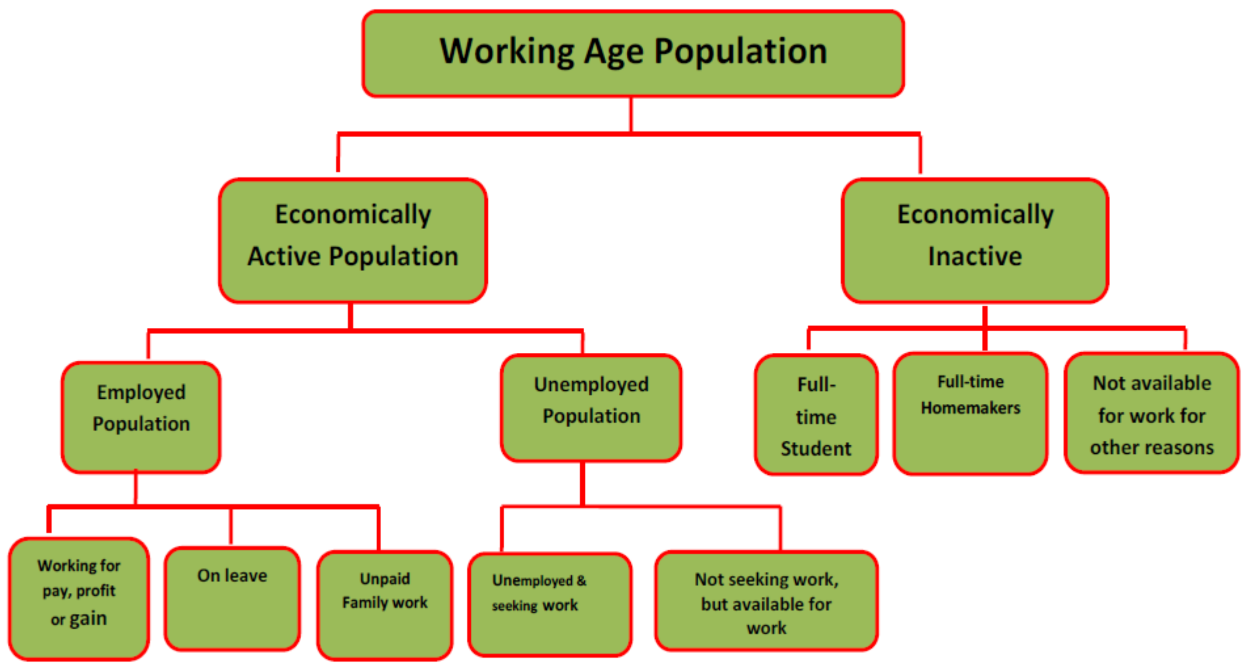
h. **Labor Force:** Sum of numbers of people **employed or unemployed** (the size of the labor force depends on the size of the population, labor market)

i. **Employment:** the number of people who work for a pay or on their own account or are unpaid family worker

j. **Unemployment:** All those out of work, able to work and looking for a job through official channels or by other means (people willing to work but are unable to find work)

k. **Working age:** The working-age population was defined as all persons 12 years and older

Figure 6.1: Organogram for the structure of Population 12 years and above



Source: 2010 Census of Population and Housing

MEASUREMENT

Labor force Participation Rate: This is ratio of the economically active population to the working age population expressed as a percent.
Unemployment rate: This is the proportion of the labor force who have no jobs, are available for work and are seeking work in a given reference period in the total labor force expressed as a percent.

Youth Unemployment Rate: This was defined as a proportion of the labor force aged 15-35 years who had no jobs, were available

SOURCES

a. Census
b. Labor force survey
c. Establishment lists, surveys
d. Employment records
e. Social security records e,g NAPSA, workers Compensations
f. Insurance companies
g. Ministry of Labor, CSO in Zambia
h. Other international data bases; US Census bureau, ILO, UN year books

OBJECTIVES OF LABOR STATICS

a. Provide a description of the size, structure and characteristics of various participants in the labor market

b. Measure changes in the employment market/labor force

c. Measures the county’s unemployment status and challenges thereof

d. Levels of unemployment over time I a given area, country or region

USES OF LABOR STATISTICS

a. Provide an adequate statistical base for the analysis of economic and social problems of employment and unemployment

b. Enables the estimation of future labor requirements

c. Crucial for estimating labor productivity

d. Enables the estimation of the purchasing power and consumption patterns

e. For man power planning and man power requirements

Limitations

a. Labor statistics in Zambia (Africa or developing country at large) are inadequate, thus an inadequate measure of economic performance due to the large informal sector

b. Differences in definitions makes international comparability difficult

    - ✓ Due to difference in time reference

- ✓ Variations of what employment, unemployment, underemployment
- ✓ Different measures used
- ✓ Differences in the age categories consisting of working population and labor force (economically active)

c. Lack of systematic, consistent mechanism of collecting, collating of labor statistics and labor issues in general

d. Unemployment is usually difficult to measure due to the irregular number of hours spent on informal works such as street vending, car watching, shoe cleaning, herding cattle, farm work, house work

3. HOUSING STATISTICS

Housing is any form or shelter used to protect persons their properties from sun, rain and or wind. Housing statistics describes the housing situation at any well-defined point in time and reflect the changes in the housing the situation at a specified period of time.

SOURCES OF HOUSING STATISTICS

- ✓ Census of Population and Housing
- ✓ Housing Institutions (Both Private and Public)- National Housing Authority, Zambia National Building Society, etc
- ✓ Construction statistics which are or can be obtained from the certificates of occupancy and permits for construction of new dwellings
- ✓ Local government and housing authority

USES OF HOUSING STATISTICS

a. Provide a systematic account of the changes in the supply of housing available to the population and hosing situation

b. There a principal indicator of the standard of living (type of housing, type of services)

c. Basis of planning and construction of houses

d. Sometimes used to estimate the national income in terms of property, home ownership, loan schemes

LIMITATIONS

a. Squatter compounds in most developing countries makes it difficult to account for as most of them are temporal, crowded, lack of services

b. Conceptual definition differences of what housing is, thus making international comparability difficult

c. The nature of housing (materials used) make it difficult to have consistent statistic on housing

d. Mobile population affect the housing statistics (Fish mongers)

e. Lack of consistent and regulated mechanism of collecting and updating housing statistics

f. Fragmented data collection mechanism by different agencies, this affects the reliability of housing statistics

4. CRIME STATISTICS

SCOPE

The Crime Rate of any country is an official statistic that reflects certain particular defining points for a country. The crime rate of a country is usually decided on the number of officially lodged cases of crime, collated over a period of usually a year.

A crime is an act contrary to the laws of the land or an unlawful act

The Crime statistics e, g (crime rate) are often considered to be an extremely important piece of information to judge the welfare and the quality/standard of living within a particular country

Crime statistics refer to statistics in a given areas at a particular timer are usually based on official criminal reports and investigation or reported, recorded and tried in the courts of law and are punishable

FACTORS AFFECTING RECORDING OF CRIME STATISTICS

a. Crime statistics often tend to overlook unofficial instances of criminal activities, and unreported/unproven crimes

b. **Geographical Location**: Rural areas are normally left out as they may not have a police station within the area

c. **Ration of the Police to the People:** The number of the police relative to the size of the population often affects crime statistics. Usually in place where the ratio is good, reports of crime tends to be good and vice-versa

d. **Discretion of the Police and other law enforcement agencies:** Petty cases often tends to end I negotiation with the law enforcement agencies and as such go unrecorded

e. **Effectiveness of the police and law enforcement:** the number and quality of crime statistics usually depend on how effective the law enforcement agencies deal with crimes reported

f. Access and Adequacy of the legal system: access to courts, delays in obtaining a ruling often weakens people’s confidence in reporting crimes

g. Traditional values and beliefs

h. Misclassification of crimes

SOURCES OF CRIMINAL STATISTICS

a. Police Departments
b. Courts of Law
c. Victim Support Units
d. Self-Reported studies
e. Anti -Corruption Commissions
f. Interpol
g. UN Law-keeping Corps

CRIME INDICATORS

a. **Crime Volume:** This is a basic indicator of the frequency of known criminal activity (relies on the number of reported cases)**.** For crimes such as murder, rape, defilement, manslaughter it represents number of victims whereas for crimes such as burglary, motor vehicle theft, arson it represents the number of known incidents
b. **Crime Rate:** this is an index of all reported crime activity standardized by population. More refined than the crude crime volume. In other words, A crime rate/Offense Rate is the number crimes/offenses per 100,000 population in a given year and area.
c. **Clearance Rate:** this is the percentage of crimes cleared
d. **Crime Trend:** This is the percentage change in crime based on data reported in a prior period. Trends can be computed for any time period such as months, quarters or years. An important measure for offence and arrest analysis/crime analysis.
e. **Population at risk rate:** This is a refined rate computed in units for the population inclined to be victimized.

USES

a. A measure of the welfare and the quality/standard of living within a particular country or region
b. A reflection of the human rights violation often if perpetrated by the state
c. Basis for examine the nature of social control and crime in a given area
d. Basis for recruiting law enforcement agencies (minimal level)

e. Examine the pattern, trends of current crimes

LIMITATIONS

a. Crime statistics are often limited to official criminal reports and investigation; hence it overlooks unofficial criminal activities

b. Every community has a unique social, ethnic and economic configuration that may affect its crime statistics, this dissimilarities may affect results for comparative analysis

c. The decision to use any indicator of crime should be exercised with great caution as no single indicator for crime analysis is a panacea for crime analysis

d. Traditional beliefs great influence crime statistics not only in recording but also in the arrests, clearing of many cases

5. AGRICULTURE STATISTICS

SCOPE

Agriculture plays a vital role in the Zambian economy. Over 70 per cent of the rural households depend on agriculture as their principal means of livelihood. Agriculture accounts for one-firth (20.2%) of the nation's Gross Domestic Product (GDP), employs about 62.8% of the labor force. In view of the predominant position of the Agricultural Sector, collection and maintenance of Agricultural Statistics assume great importance.

The statistics collected attempts to measure among other things:

- ✓ How many people are working in agriculture
- ✓ Crop production
- ✓ Crop area production
- ✓ Inputs
- ✓ Land area for crop production
- ✓ Forestry
- ✓ Livestock products
- ✓ Importance of agriculture in economic growth, its economic share of GDP
- ✓ Market prices of agriculture products
- ✓ Food security (both household and national)
- ✓ Characteristics of people engaged in agriculture

SOURCES OF AGRICULTURE STATISTICS

- ✓ Census of population and Housing
  The census tries to provide information on:
  - o Number and characteristics of people engaged in agriculture and other economic activities
  - o Area planted (type of crops, trees etc)
  - o Average yields and production of crops (maize, cassava, millet, soya beans, sun flower, cotton, tobacco etc)
  - o Number of agriculture households
  - o Data on farming patterns (inputs, equipment etc)
- ✓ Living Condition Monitoring survey
- ✓ Agriculture census
- ✓ Rural household surveys
- ✓ Local Agriculture Corporative

USES OF AGRICULTURE STATISTICS

- ✓ Assess the food security of a nation
- ✓ Determine the provision of farming inputs, equipment and which population requires assistance in the production of crops
- ✓ Basis of measuring the economic performance, well-being of the population
- ✓ An indicator of industrialization
- ✓ Agriculture statistics forms the basis for conducting agriculture surveys
- ✓ Classification of the population (peasant, commercial farming)

LIMITATIONS

- ✓ Limited coverage (most agriculture producers are peasant farmers)
- ✓ Timeliness of data
- ✓ Comparability
- ✓ Lack of consistent and systematic mechanism of collecting agriculture statistics (Most African countries do not have a robust system of collecting agriculture statistics)

6. GENDER STATISTICS

SCOPE

Gender statistics is an area that cuts across traditional fields of statistics to identify, produce and disseminate statistics that reflect the realities of the lives of women and men, and policy issues relating to gender.

Statistics and indicators on the situation of women and men are needed to describe the role of women and men in the society, economy and family, formulate and monitor policies and plans, monitor changes, and inform the public.

Gender statistics incorporate issues on roles; gender roles are learned expectations and behaviors in a given society which are perceived male or female. Some roles cut across political, economic, and geographical boundaries.

The gender analysis is anchored on some questions, and these form the core for gender analysis:

i. Who does what?
ii. Who owns what?
iii. Who has access/controls what?
iv. Who knows what?
v. Who benefits what?
vi. Who should be included in economic and development programs, key decision making?

These questions can also help in identifying gender specific indicators used in many sectors.

GENDER INDICATORS

Examples of Core indicators used in Gender analysis

- ✓ Percentage of holdings/households by sex of the holder/household head
- ✓ Average age of the holder/household head and household members by sex of holder/household head
- ✓ Percentage of holdings/households with holder/household head with education level over a CERTAIN level by sex
- ✓ Percentage of holdings/households receiving agricultural extension services by sources of agricultural extension services and sex of holder/household head
- ✓ Percentage of holdings/households with irrigated land by land use type and sex of holder/household head
- ✓ Percentage of holdings/households by type of farming (crop (temporary, permanent), livestock, aquaculture and forestry) and sex of the holder/household head
- ✓ Percentage of holdings/households receiving credit for agricultural purposes by sex of holder/household head

- ✓ Percentage of women in key decision making
- ✓ Ratio of women to men in government

Other important indicators include:
The Gender related Development Index (GDI): The GDI adjusts the Human Development Index (HDI) for gender inequalities in the three dimensions covered by the Human Development Index (HDI), i.e. life expectancy, education, and income. It is important to note that the GDI is not specifically.

The Gender Empowerment Measure (GEM): GEM seeks to measure relative female representation in economic and political power. It considers gender gaps in political representation, in professional and management positions in the economy, as well as gender gaps in incomes.

Gender Equity Index (GEI): GEI is tries to measure gender rating I three dimensions education, economic participation and empowerment or it is a composite index that measures or captures the loss of achievement between the sexes within a country, due to gender inequality, and uses these three dimensions to do so. It makes it possible to classify countries and rank them in accordance with a selection of gender inequity indicators in three dimensions;

- ✓ **Education:** measured by the literacy gap between men and women and by male and female enrolment rates in primary, secondary and tertiary education.
- ✓ **Participation in the economy:** measured by the percentage of women and men in paid jobs, excluding agriculture, and by the income ratio of men to women.
- ✓ **Empowerment**: measured by the percentage of women in professional, technical, managerial and administrative jobs, and by the number of seats women have in parliament and in decision-making ministerial posts.

TYPES OF GENDER STATISTICS

Most gender statistics are biased towards women;
- ✓ Gender based violence
- ✓ Divorce initiated by women
- ✓ Reproductive rights
- ✓ Participation of women in key decision making

SOURCES OF GENDER STATISTICS

- ✓ National Surveys (Zambia Demographic Health
- ✓ National or International publications
- ✓ Administrative Records e.g Zambia Police (Victim Support Unit)
- ✓ Government Ministries (Ministry of Gender, Community Development Mother and child Health)
- ✓ NGOs, Civil organizations

- ✓ UN Statistics Division
- ✓

LIMITATIONS

- ✓ Quite difficult to compile as they are a new form of statistics in Africa
- ✓ Culture and traditional believes underplays the importance of gender statistics
- ✓ Lack of existing data on gender makes it difficult to put up a stimulated system of collecting ender statistics
- ✓ Lack of systematic data collection on gender inequality and especially on violence against women that is considered always a private issue not to be reported is a large problem.

PRESENTATION OF SOCIAL, ECONOMIC AND DEMOGRAPHIC DATA

Basic Measures (Recap)

1. **RATIO**

   The Quantitative relation between two amounts showing the number of times one value contains or is contained within the other. Mathematically, a ratio shows the relative sizes of two or more values.

   Examples of ratios include;

   a) Sex Ratio- The number of males per 100 females
      SR= (M/F) *100
   b) Dependency Ratio- The number of dependents per 100 economically active population

$$DependencyRatio = \frac{Population(0-14) + Population65+}{Population15-65} \times 100$$

   Other examples

   Doctor - patient ratio, Teacher-pupil ratio

2. **PROPORTION**

   A proportion is a part or share of a whole; it is a ratio which expresses the relative size of a number in terms of total. It can also be defined as a part, share or number considered in comparative relation to a whole.
   Examples include;
   Femininity ratio, Masculinity ratio, Proportion of HIV+ persons eligible for ART initiation.

3. **PERCENTAGE**

   A proportion expressed as a base of two or more numbers. The constant used is 100 for easy interpretation and comparison.
   Examples include:

   % of total population with access to education
   % of total population with access to Health
   % of total population with access to clean water

4. **RATE**

   A ratio of one figure to another in a specified period of time.

   Examples Include;

   Pregnancy Rate = Number of pregnancies/ Total Population of women(15-49)* 1000

Sometime a rate is referred to as an occurrence or exposure ratio because the numerator is a number of occurrences or event usually in a year while the denominator measures the number of people exposed to the risk of experiencing the event.

METHODS OF DATA PRESENTATION

Recap! Types of Data

Types of Data

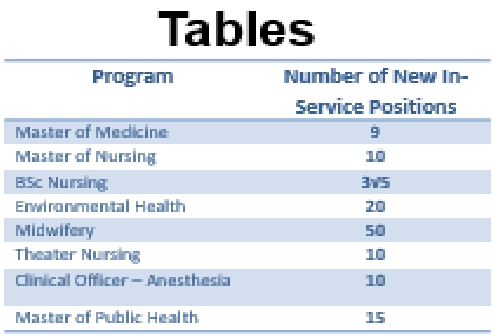
1. Continuous Data: These are numerical data that are measured in an unbroken scale such as age, weight and CD4 count. Often these data can be categorized during data analysis.
   - Age: . . .18, 19, 20, 21, 22, 23 . . .
   - Weight: . . .150, 151, 152, 153 . . .
   - Temperature: … 37, 37.2, 37.9, 38…
2. Categorical: These are data that can be broken into distinct categories, such as gender and marital status.
   - Age Categories: 18-24, 25-44, 45-64, 65+
   - Sex: Male, Female
   - HIV status: positive, negative

Data cleaning: Data cleaning is the first step in data analysis and involves looking for obvious errors in data entry that can be followed upon and fixed.

Why do we need to present data?

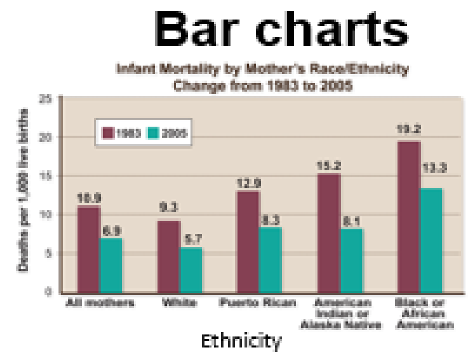
- Communicate with decision makers
- Document events
- Demonstrate successes or failure
- Show trade-offs between 2 choices
- Make decisions

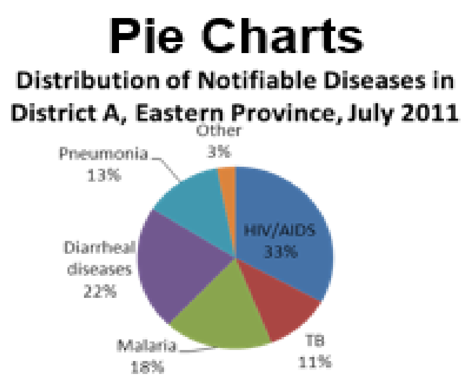
Common ways to display data – tables and charts

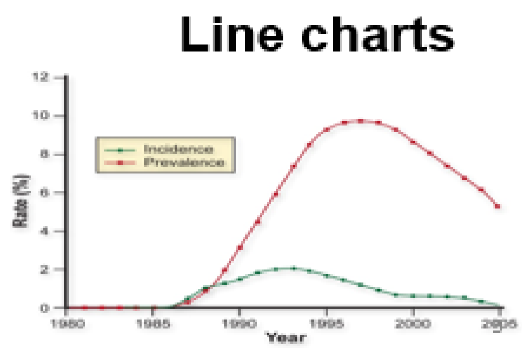


Tables

Program	Number of New In-Service Positions
Master of Medicine	9
Master of Nursing	10
BSc Nursing	3VS
Environmental Health	20
Midwifery	50
Theater Nursing	10
Clinical Officer – Anesthesia	10
Master of Public Health	15







1. TABLE

A table is a means of arranging data in rows and columns. It is a device used to summarize information.

A table might not be so appropriate for displaying data especially if the number of rows and columns is too large. Such data has to be summarized on a graph.

Table 1.0: % of HIV Cases by Sex, Province Y

Title

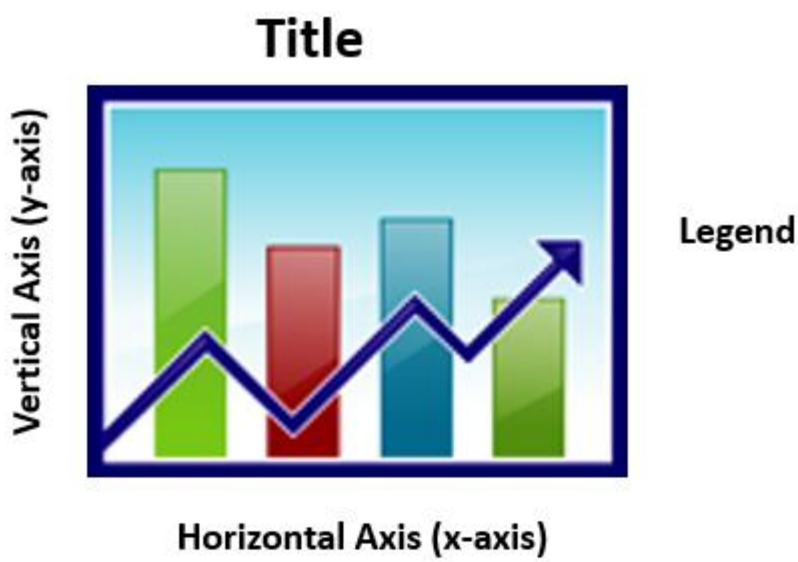
District	Female	Male
A	90	10
B	85	15
C	70	30
D	90	10
E	50	50
F	50	50
G	60	40
H	90	10
I	80	20
J	60	40

Row

Column

Tables are useful for data presentation; however, they take time to read.

CHART



GENERAL GUIDELINE

- Decide on the one MESSAGE you want people to remember from your graphic
- Delete every piece of information, animation, color, dimension or shape that could distract from this MESSAGE

CHARTS can

1. Summarize and display complex data clearly and effectively and emphasize specific points
2. Identify and present distributions, trends, and relationships among the data
3. Easily communicates findings for planning groups and decision makers

However, poorly designed and executed tables and figures can mislead users or distract them from your message.

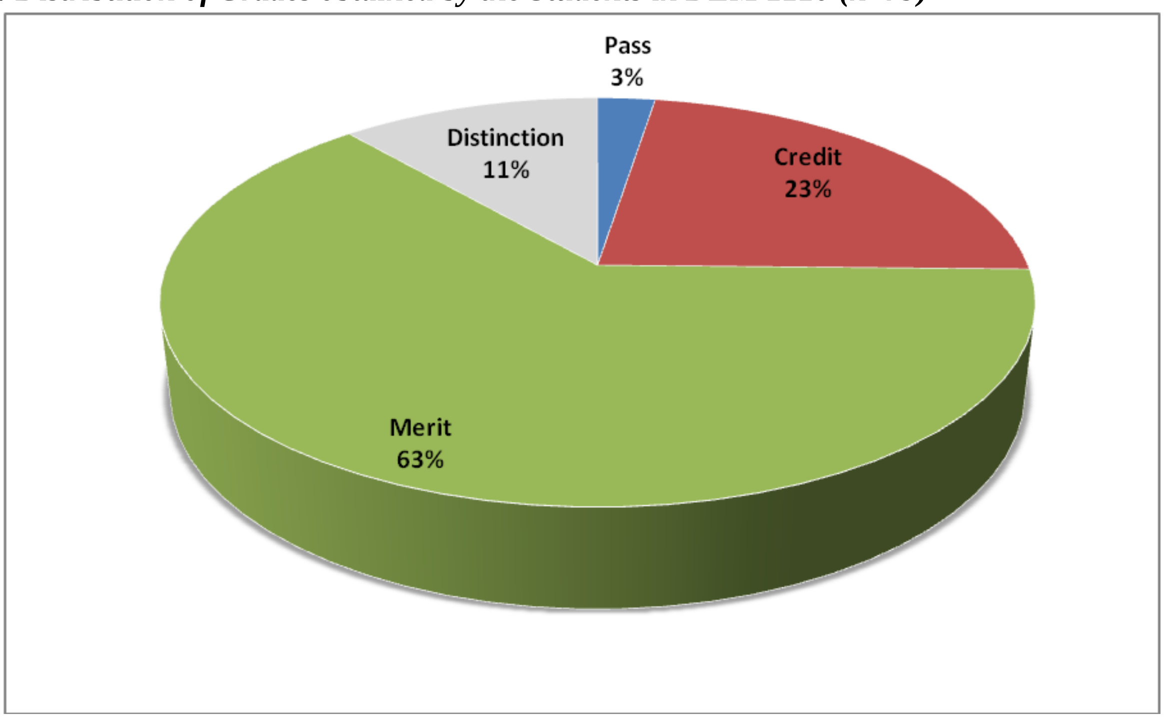
Key points to consider when using charts to display data

1. What information is being conveyed in each?
2. What different points do they make?
3. Is this the most appropriate chart type to represent these data.

1. PIE CHART

- Circular chart split into segments which show components of a larger group
- Suitable for displaying categorical data, or discrete data in distinct categories
- The size of a “slice” is proportional to the amount of data (e.g., number of cases) it represents
- Proportion (%) of the total each component represents is frequently added on the slice
- Different colors can be used to identify various slices

Figure: Distribution of Grades obtained by the Students in DEM 1110 (n=79)

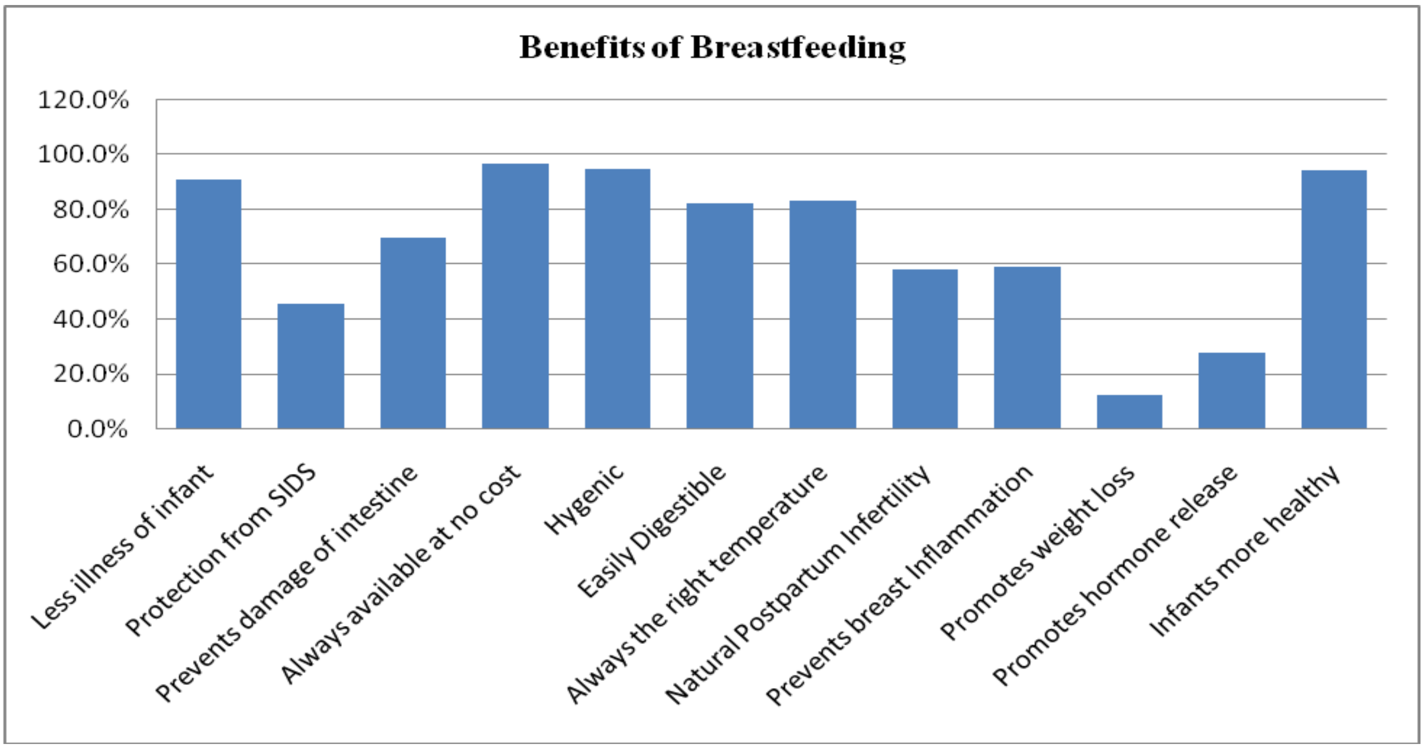


WHEN NOT TO USE A PIE CHART

- When you would have more than 8 slices
- When the values of each slice are similar because it is difficult to see differences between slice sizes
- When you have a limited number of cases or total (e.g., only two cases of measles out of 12 people)
- To compare with another pie chart

2. BAR CHARTS

- Display categorical data and compare discrete data in distinct categories
- Each bar represents one category
- Can be organized horizontally or vertically
- Height of the bars are proportional to the number of events (e.g. cases) in the category
- Data in a bar chart can be discrete (e.g. sex, region, race) or continuous (e.g. age) but organized in categories (e.g. age groups)

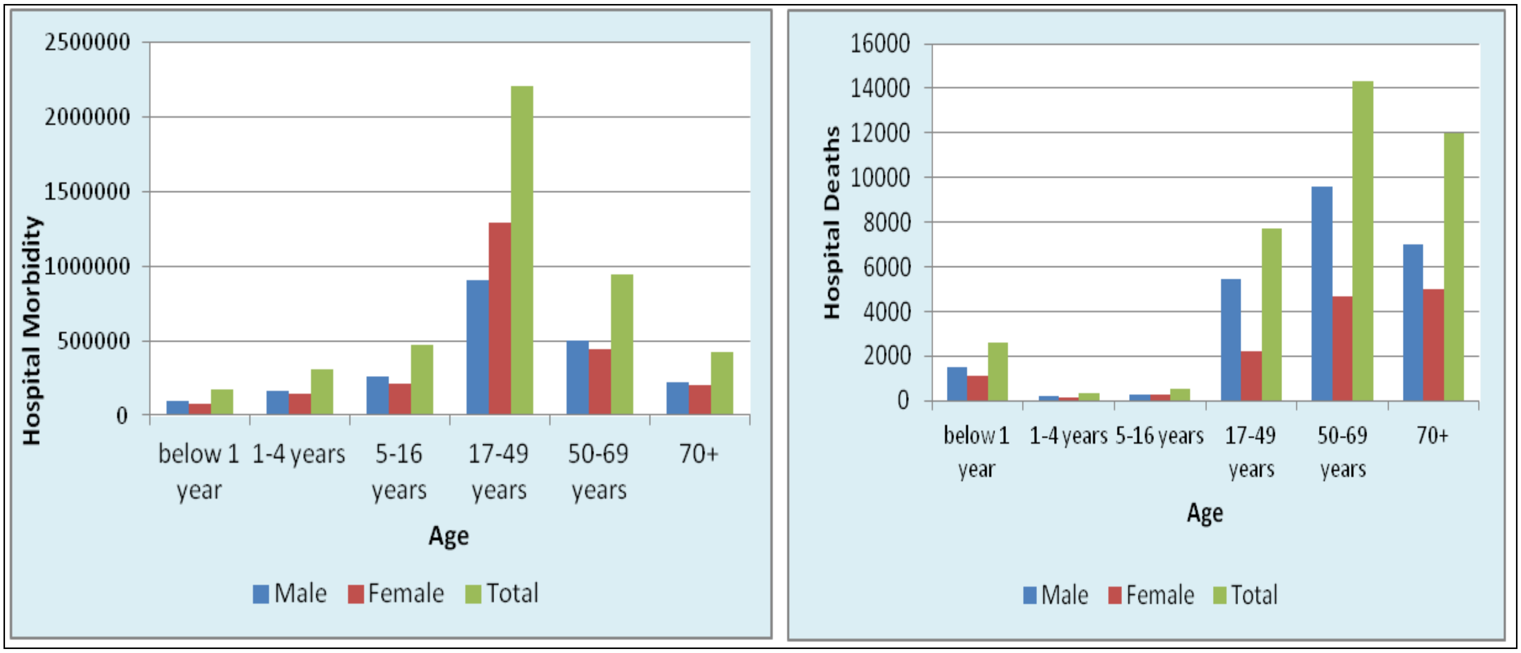


Three common types of bar charts

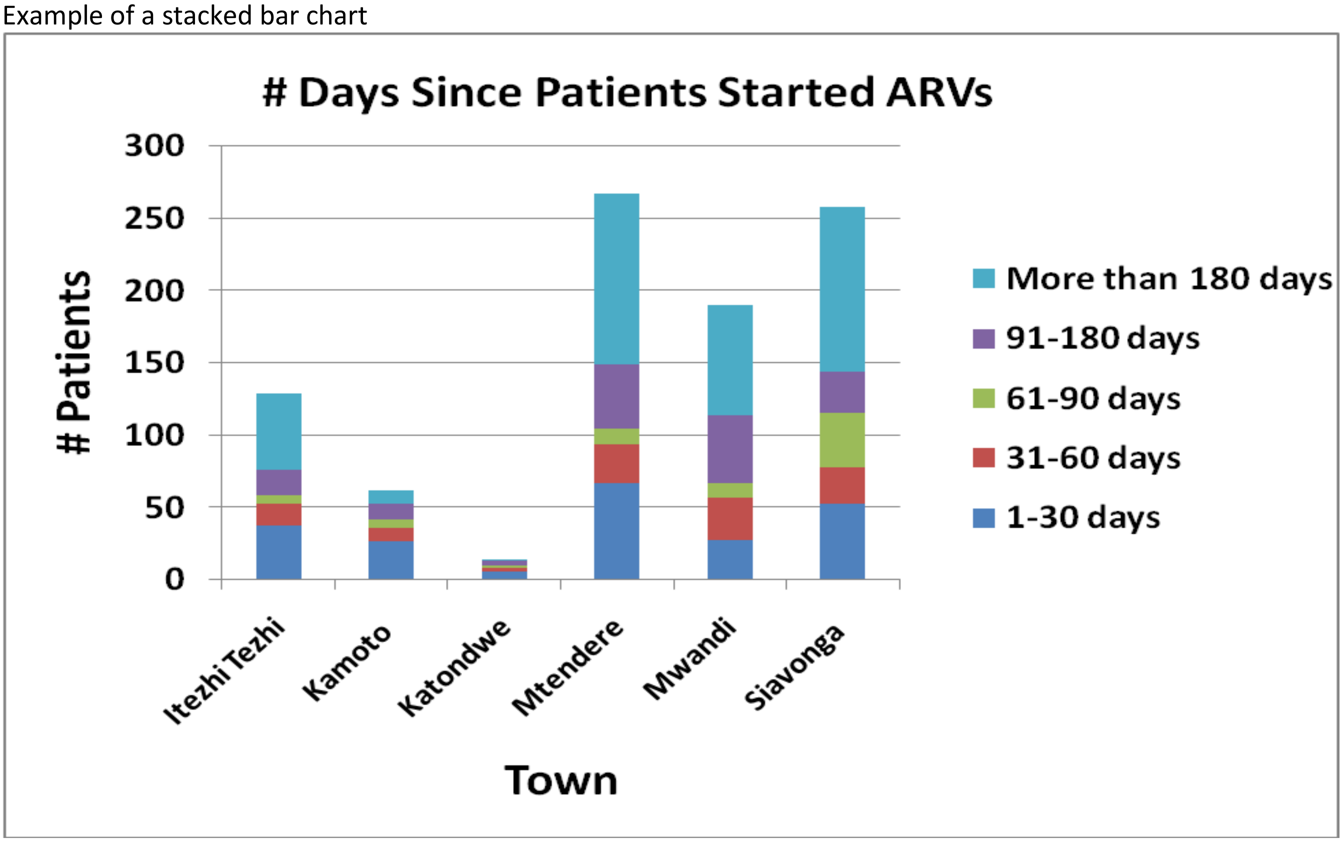
1) Simple
2) Grouped
3) Stacked

Example of grouped bar chart

Figure: Distribution of Hospital morbidity and Hospital deaths in year by Age



Example of a stacked bar chart



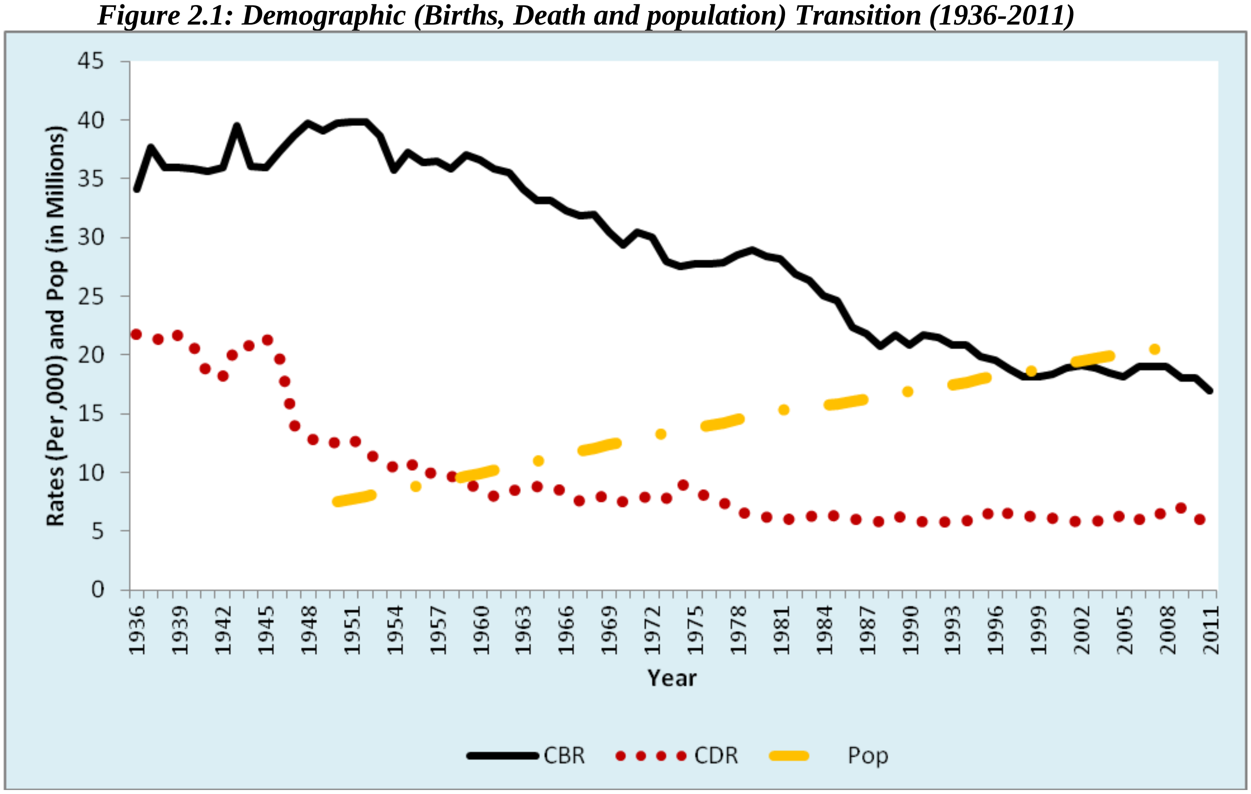
3. LINE CHART

- Used for continuous variables
- Especially useful to convey passing time
- Display relationships between 2 variables
- When used appropriately, can summarize and display complex data clearly and effectively and emphasize specific points
- Let you identify and present distributions, trends, and relationships among the data

Example of a line graph

Figure1 : Number of Clinicians Working in Each Clinic During Years 1–4*

Figure 2.1: Demographic (Births, Death and population) Transition (1936-2011)



Source of Data: US Census Bereau Website (http://www.census.gov/population/international/data/idb/estandproj.php

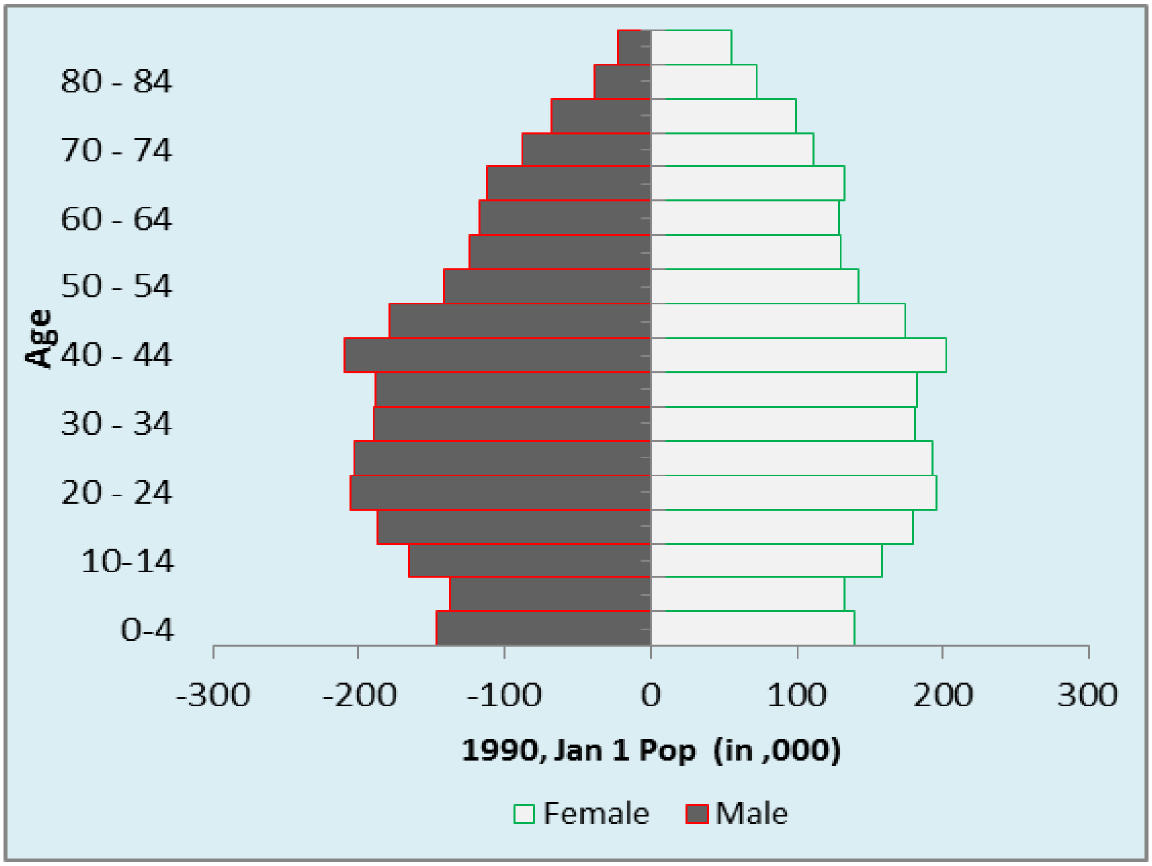
Limitations of line charts

- Dependent on accurate data input from tables and figures
- Can be difficult to see actual numbers, and limited to the number of lines in a chart

4. POPULATION PYRAMID

A population pyramid is a graphical illustration that shows the distribution of various age groups in a population. It is used to depict the age-sex structure of a population

Example of population pyramid: Egypt population pyramid 2005



SUMMARISING CONTINOUS DATA

- Mean
- Median
- Mode
- Range

1. **Mean**

Mean = <u>sum of value</u>
# of observations
(1,2,3,4,5)
Mean = 1+2+3+4+5/5

Adding an outlier changes the mean dramatically! Use means only when data are normally distributed – with no outliers

2. **MODE**

Observation that occurs most frequently

9 12 15 15 15 16 16 20 26

Mode =15

3. **Median**

Middle value when observations are arranged in order of magnitude.

4. **Range**

The spread of the data from lowest <u>(minimum)</u> value to highest value <u>(maximum)</u>

Example:
9 12 15 15 15 16 16 20 26
Range = (9—26)

EVALUATION OF DEMOGRAPHIC DATA

Demographic data are used in various aspects that encompass all aspects of human, economic, social and political development. Various sources are used to capture social, economic and demographic data such as the census, surveys, vital registration systems and population registers among others. In developing countries the common methods are the censuses and surveys. Despite the surveys and census being the dependable sources of data, their reliability various across countries and source to source. Thus, social, economic and demographic data need to be evaluated before it is used.

Why evaluate data (social economic and Demographic data)

a. To identify sources and types of errors in order to know which sections of data contain errors and what procedures of data collection and processing lead to these error, useful in planning for future preventive measures
b. To determine the accuracy of data collected
c. To make appropriate adjustment to the data before we can use confidently to estimate demographic events, plan for social and economic development.

Types of errors

There are basically two types of errors;

a. Coverage Errors

These are as a result of either under enumeration/over enumeration of a place, village, or persons

b. Content Errors

These arise as a result of;

- ✓ Interviewer biases
- ✓ Use of wrong questionnaires
- ✓ Misunderstanding of questions, memory lapse, deliberately giving wrong answers
- ✓ Data entry

Principle Sources of Errors

- ✓ Errors resulting from inadequate preparation in data collection, collation, analysis
- ✓ Errors committed during data collection
- ✓ Errors due to processing of data (editing, coding, transcription, entry)
- ✓ Sampling errors

METHODS OF DETECTING ERRORS

There are basically two methods of evaluating data;

a. **Direct Methods**

This method entails having two independent methods that should collect the same information e.g the census and the post enumeration survey. The post enumeration survey is done on a selected area. The results of the post enumeration survey are compared with the results of the census in that selected area.

b. Indirect Methods

- ✓ **External consistency** check method (Involves comparing the results from one census to another or the census with other sources e.g survey, VRS
    - i. Balancing equation
    - ii. Error of closure
    - iii. Intercensal growth rate
- ✓ **Internal Consistence Check;** this method rely heavily on age sex data, this is because age and sex influence a lot of social, economic and demographic data.
    - i. **Single year age;** this method detects errors as a result of age misstatement, age preference in numbers ending with 0, 5 even numbers. Under this method the following are used;
        - Graphical method which provides a visual inspection of the data
        - Mathematical methods which involves methods of deriving indices f digit preference such as;

1. Index of age preference

- ✓ **Whipples index;** This method was developed to detect the degree of age misstatement that can be attributed to the preference of age ending in digit 0 and 5
- ✓ **Myers Blended index;** this method was developed to detect the preference of all digits ending 0 to 9. This method gives equal weight in terms digit preference in all the digits.

Methods of Smoothening/Graduating data

- o Graphical methods
- o Mathematical methods
    - UN Method
    - Carrier-Farag method